

An optogenetic smart microscopy platform reveals signaling dynamics-dependent control over collective cell migration

Highlights

- PyCLM enables the easy design of complex optogenetic stimulation experiments
- PyCLM supports embryo patterning, biosensor feedback control, and tissue migration
- We apply PyCLM to studying EGFR-dependent tissue migration in MDCK monolayers
- Fast versus slow EGFR activity waves can elicit different directions of tissue migration

Authors

Harrison R. Oatman, Beena C. Lad, Long Nguyen, Justin H. Gubbens, Jared E. Toettcher

Correspondence

toettcher@princeton.edu

In brief

Oatman et al. introduce PyCLM, a Python suite for closed-loop optogenetic experiments combining real-time biosensor measurement and segmentation of thousands of cells. The method is applied to epithelial migration, where single-cell and tissue-scale control of light-activated EGFR signaling reveals that a critical wave speed reverses migration direction.

Report

An optogenetic smart microscopy platform reveals signaling dynamics-dependent control over collective cell migration

Harrison R. Oatman,¹ Beena C. Lad,² Long Nguyen,³ Justin H. Gubbens,⁴ and Jared E. Toettcher^{2,3,5,*}

¹Lewis Sigler Institute, Princeton University, Princeton, NJ 08540, USA

²Department of Molecular Biology, Princeton University, Princeton, NJ 08540, USA

³Omnenn-Darling Bioengineering Institute, Princeton University, Princeton, NJ 08540, USA

⁴Khoury College of Computer Sciences, Northeastern University, Boston, MA 02115, USA

⁵Lead contact

*Correspondence: toettcher@princeton.edu

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SUMMARY

In cell biology, optical techniques can measure cells' internal states (biosensors) and stimulate cellular responses (optogenetics). Yet the design of all-optical experiments is often manual: a predetermined stimulus pattern is applied to cells, biosensors are measured over time, and data are processed offline. Here, we develop PyCLM, a Python-based suite enabling closed-loop measurement, image segmentation, and optogenetic control of thousands of cells per experiment. We showcase PyCLM on diverse applications, including performing feedback control on single cells and delivering developmental signaling patterns to *Drosophila* embryos. We compare single-cell versus tissue-scale optogenetic control of epithelial migration, revealing that fast and slow waves of receptor tyrosine kinase activity determine the direction of tissue movement, matching prior *in vivo* observations in zebrafish and mouse. PyCLM enables simple setup of dynamic experiments to probe cell and tissue properties and provides a first step toward real-time control of single-cell states at the tissue scale.

INTRODUCTION

Optical techniques play increasingly important roles in modern cell biology. Sophisticated fluorescence imaging techniques continue to push the limits of speed, resolution, and 3D volumetric acquisition.^{1–3} Meanwhile, live-cell biosensors are available for a large number of cellular processes, from kinase activity to metabolism,⁴ and optogenetic tools can now be used to activate protein binding, phase separation, cell signaling, and gene expression.⁵ Rapid advances in artificial intelligence and machine learning for image processing have also made it possible to extract increasingly sophisticated information from biological imaging data,⁶ opening the door to studying the behavior of large numbers of cells in complex geometries.

The increasing availability of optogenetic actuators, fluorescent biosensors, and computational algorithms makes it possible to envision automating live-cell microscopy experiments that combine real-time measurement and rule-based stimulation. Experimental automation is particularly needed in the case of dynamic, rare, or collective biological phenomena,^{7–9} where it rapidly becomes impractical or impossible to manually identify cells of interest and target them with individualized stimuli or to simultaneously stimulate enough cells to initiate a collective tissue-scale response.

A growing number of “smart microscopy” approaches have been recently introduced to address these open challenges. Smart microscopy refers broadly to an emerging set of adaptive techniques in imaging and stimulation, empowering a microscope to autonomously decide how to image or stimulate a biological sample.^{10–12} Khammash and colleagues implemented a low-cost hardware and software setup for imaging and optogenetic stimulation, enabling virtual computer players to stimulate fields of cells in a game of tic-tac-toe.¹³ Kapitein and colleagues recently reported a system for performing real-time control over nucleocytoplasmic transport and movement of individual cells.¹⁴ Pertz and colleagues extended image-based optogenetic stimulation across scales, from subcellular stimulation in single cells to multiple stimuli across populations of cells.¹⁵ These approaches are very powerful, yet each requires considerable programming by the end user to implement customized experiments, particularly when different experiments are to be performed in parallel on different fields of cells. We reasoned that an ideal system should be rapidly reconfigurable for many different tasks, from single-cell control to tissue-scale stimulation, and should be as simple to use as possible, without the need for any programming by the end user to combine any set of previously defined workflows into a new experiment.

Here, we have developed a software package that accomplishes these goals called Closed-Loop Microscopy in Python

(PyCLM). PyCLM combines fluorescent image collection through MicroManager,¹⁶ cell segmentation through Cellpose,¹⁷ and control of a digital micromirror device for optogenetic stimulation. Our system can be used to deliver dynamic light inputs to multiple fields of view, stimulate large numbers of cells, and perform closed-loop experiments where real-time results of cell segmentation and biosensor activity are used to automatically update light inputs delivered to each cell. Any of these experimental modalities can be combined and configured independently for different fields of view during a single multi-point microscopy session using simple text-based configuration files, without end-user programming. We showcase PyCLM across a variety of applications, including implementing single-cell feedback control to simultaneously regulate the fluorescence intensity of $\sim 1,000$ cells in an imaging field, applying scaled and rotated developmental patterns to randomly oriented *Drosophila* embryos, and delivering tailored subcellular light stimuli to an optogenetic epidermal growth factor receptor (OptoEGFR) to drive directed tissue movement.

We further apply PyCLM to address an open biological question: how is tissue migration organized by spatiotemporal dynamics of receptor tyrosine activity? Recent studies have revealed that developmental and regenerative processes in epithelial tissues are often accompanied by propagating waves of receptor tyrosine kinase (RTK) signaling. Waves of fibroblast growth factor (FGFR) signaling travel at ~ 10 $\mu\text{m}/\text{h}$ across regenerating zebrafish scales,¹⁸ whereas ~ 100 $\mu\text{m}/\text{h}$ waves of EGFR activity are prominent in many mammalian epithelial tissues.^{18–20} It has been hypothesized that these traveling waves of RTK signaling serve to organize collective cell movement toward sites of tissue damage. Applying our PyCLM workflow to epithelial cells expressing a light-activated EGFR, we find that tissue-scale waves of RTK activity are sufficient to induce rapid and controllable tissue-scale migration. Our experiments reveal two qualitatively different modes of tissue movement that depend on the speed of RTK activity waves: slow (15 $\mu\text{m}/\text{h}$) and fast (60 $\mu\text{m}/\text{h}$) light waves produce opposite directions of tissue movement, matching observations during zebrafish scale regeneration and mouse epithelial wound healing. Overall, we anticipate that the PyCLM toolbox will be broadly useful for probing dynamic single-cell biology, delivering developmental inputs, and examining collective cell behaviors.

RESULTS

The PyCLM architecture for closed-loop imaging and optogenetic stimulation

Our overall goal was to develop software to enable complex imaging and stimulation experiments for cell and developmental biology (Figure 1A). Current optogenetic experiments often require the ability to deliver dynamic light patterns, such as traveling waves or light gradients, for the study of tissue migration or developmental patterning,^{21,22} yet specification of these dynamic patterns remains difficult in standard software packages. We also sought to integrate real-time image segmentation and light stimulation so that the location and intensity of stimulation could be tailored to the geometry or activity state of each biological sample. Such “closed-loop” experiments allow light stimulation to adaptively respond in a programmatically predetermined way as an experiment progresses.

Our design of PyCLM was guided by several practical design considerations. First, we sought to achieve a high degree of flexibility so that many different experimental workflows could be performed on different fields of view in a single imaging session. Next, we aimed to build an extensible system so that diverse experiments could be designed and implemented with ease. For example, one must be able to swap in and out different cell segmentation models, apply different time-varying light input patterns to each field of view, or implement rules for context-aware stimulation (e.g., feedback controller architectures). Finally, we sought a simple interface so that users without a programming background could remix any existing combination of modules to define new experiments. To produce such an accessible and generalizable workflow, we brought together resources that are already used in a diverse set of experimental systems and image processing tasks: Cellpose¹⁷ for generalist image segmentation, and MicroManager¹⁶ for microscope control and optogenetic stimulation.

Scheduling can pose a major challenge for smart microscopy implementations. To satisfy the closed-loop design, image acquisition, cell segmentation, and optogenetic stimulation must all be performed in real time before the next round of imaging. Our solution to this challenge is an architecture where processes (imaging, cell segmentation, and stimulation) are implemented as distinct modules that run as independent threads (Figure 1B). A master scheduler process manages each of the modules to ensure appropriate sequencing of events. We implemented a scheduler and five classes of modules: (1) the “microscope manager” that works with MicroManager to control stage movement, image acquisition, and optogenetic stimulation; (2) the “microscope outbox” that controls data storage and handles images; (3) the “segmentation module” that implements image segmentation; (4) the “pattern module” that defines any dynamic illumination patterns or updating rules (e.g., feedback control); and (5) the “pattern manager” that stores patterns and provides them to the microscope manager as needed. The user may configure one of these modules—the microscope manager—to interface with their hardware. Specifying segmentation and pattern methods can then be used to implement different experimental designs. Each field of view uses its own specific combination of modules and configuration parameters so that many different experiments can be run in parallel during a single multi-point acquisition.

From the end-user perspective, running a PyCLM workflow is performed not by writing Python code but by editing a set of human-readable Tom’s Obvious, Minimal Language (TOML) configuration files and providing an XML or MicroManager .POS file containing a list of stage positions associated with each experiment type. The .TOML configuration files allow the user to set any parameters needed for imaging, segmentation, and optical stimulation. Each stage position is associated with its own .TOML file that describes the specific experiment to be performed at that position, and a master schedule.toml file sets the overall experiment length and imaging frequency. The user thus needs to provide a set of imaging coordinates, a .TOML configuration file for each position, and a scheduler configuration file to specify a complete experiment (see examples of this code in Figure 1C). We provide .TOML configuration files for running all the experiments described in this manuscript.

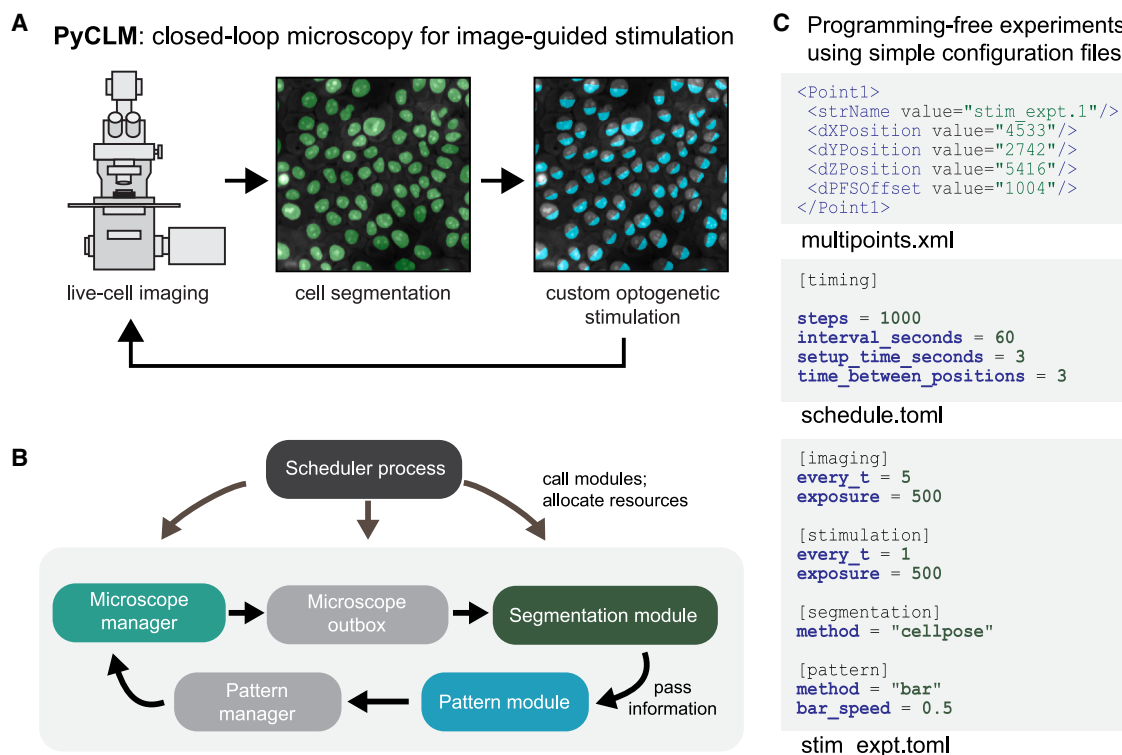


Figure 1. Overview of the PyCLM closed-loop microscopy pipeline

(A) PyCLM is designed to combine imaging, cell segmentation, and delivery of customized optogenetic patterns to many cells across multiple fields of view. (B) Organization of the PyCLM software. A master scheduler controls the sequencing of all modules. Five modules pass information in a feedback loop: a microscope manager for hardware control, an “outbox” for image data storage, a segmentation module for identifying biological structures, a pattern module for implementing rules for optogenetic stimulation, and a pattern manager that collects patterns and delivers them back to the hardware manager. Multiple microscope managers, segmentation, and pattern configurations can be combined in a single experiment to run on different fields of view and can be easily interchanged to implement different experimental designs (e.g., segmentation algorithms or stimulus designs, including dynamic light patterns or feedback-controlled illumination rules). (C) An example of the configuration files required to run an experiment. A multiPoints.xml file provides the names and spatial coordinates of each imaging position. Each position’s name is linked to a .TOML configuration file that defines a combination of modules and their configurations so that multiple complex experiments can be run in parallel. The schedule.toml configuration file specifies overall settings for the entire experiment.

and describe how to implement new pattern modules, the major user-defined component of a new PyCLM experiment (see Method S1).

Parallelizing rule-based stimulation using PyCLM

We showcase a few different use cases for PyCLM that are representative of some common challenges in optogenetics and cell biology. One class of problems involves adjusting the optogenetic inputs delivered to each cell based on its current state (Figure 2A). This can be necessary when delivering stimuli to cells in response to a rare and asynchronous event (e.g., stimulating only mitotic cells or stimulating only cells with high current biosensor activity) or may be useful to implement optogenetic feedback control to reduce cell-to-cell variability in signaling state²³ or gene expression.²⁴ Rule-based optogenetic stimulation can in principle be used to deliver tailored inputs to each cell in response to a change in state (triggered stimulation) or to eliminate intrinsic differences in gene expression or cell state (feedback control).^{25,26} However, a challenge arises at scaling up this kind of control. For large populations of cells, each cell must be segmented, its state measured, and an appropriate light

input must be delivered to each cell based on the appropriate rule.

We developed PyCLM segmentation and pattern methods that enable rule-based stimulation for large populations of single cells. To enable local stimulation of all cells in a field of view, we implemented a segmentation method that uses any pre-trained or custom Cellpose segmentation model. All of the results below use the built-in CPSAM model,²⁷ which we found to work well across a wide variety of contexts. As an example of a rule-based pattern method, we implemented a “bang-bang” feedback controller that compares each cell’s fluorescence intensity to a target value and stimulates that cell with a light input if the intensity is lower than the target value. Bang-bang control is a widely used feedback control strategy that is well known from home thermostats and has been previously used for optogenetic control over gene expression in *S. cerevisiae*.²⁸

As a test case for biological control, we cultured MCF10A cells that expressed a TagRFP-H2B nuclear marker at variable expression levels.²⁹ This system provides an ideal case for blue light-dependent control due to the fact that like many red fluorescent proteins, TagRFP is photochromic, exhibiting a

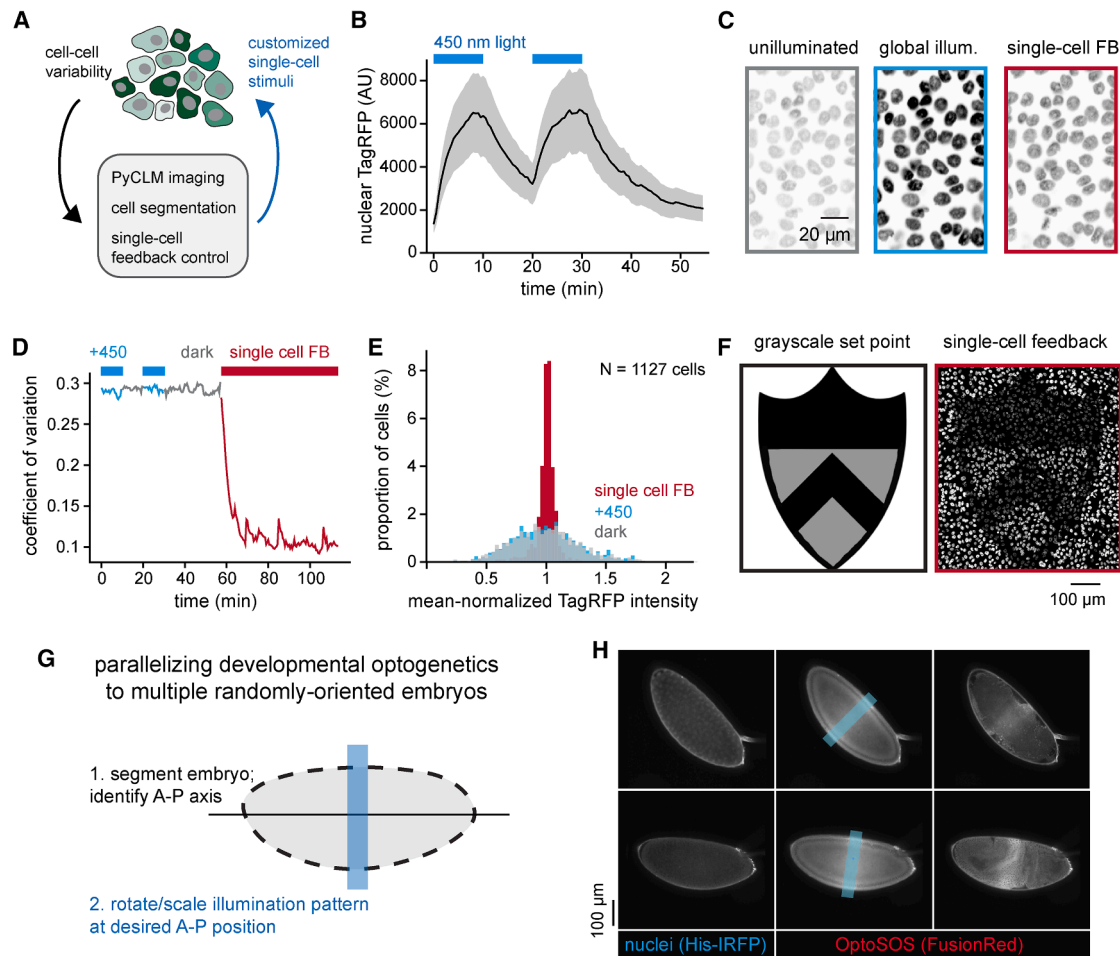


Figure 2. PyCLM delivers rule-based light stimulation using real-time imaging data

(A) Because it integrates real-time analysis and optogenetic stimulation, PyCLM can be used for real-time feedback control of cellular properties. We envisioned an experiment where heterogeneous fluorescent signals could be “normalized” by tailoring light inputs delivered to each cell.

(B) MCF10A cells expressing a nuclear TagRFP-nuclear localization sequence (NLS) construct were exposed to blue light or darkness, and their red fluorescence was measured over time. A 4-fold change in TagRFP fluorescence is observed between the absence and presence of blue light due to the photochromicity of this fluorescent protein. Curves show mean $\pm$ SD from $n = 1,127$ cells.

(C) Images from a field of TagRFP-expressing cells in the dark, under global blue light, or subjected to a bang-bang feedback controller that applies a blue light stimulus to each cell if its intensity falls below a set point. Darker colors correspond to higher fluorescence intensities.

(D) The CV in TagRFP fluorescence calculated between $n = 1,127$ cells in the field of view is plotted as a function of time under conditions of darkness, global blue light, or feedback control.

(E) Histogram of nuclear TagRFP fluorescence intensities in $n = 1,127$ cells in dark, light, and under feedback control.

(F) Feedback control to produce a spatial pattern of TagRFP fluorescence based on a grayscale image of set points. Left: the set point image. Right: the same cells after feedback control is applied.

(G) PyCLM can segment *Drosophila* embryos to apply precisely oriented patterns along the anterior-posterior (AP) axis.

(H) OptoSOS embryos before and after light stimulation. A fluorescent histone (His-IRFP; left) is imaged to monitor the developmental stage. FusionRed-tagged OptoSOS imaging reveals local membrane recruitment (middle) and tissue invagination (right) induced by optogenetic stimulation (blue overlay).

See also [Figure S1](#).

change in red fluorescence after blue light exposure.^{30,31} Performing time-lapse imaging of MCF10A TagRFP-H2B cells revealed that their nuclear fluorescence increased by approximately 4.5-fold following illumination by 450 nm light. This change is reversible upon removal of the blue light stimulus ([Figure 2B](#); [Video S1](#)). We reasoned that blue light stimuli could be tailored to each cell to overcome the initial cell-to-cell heterogeneity in TagRFP expression, producing tightly controlled cellular fluorescence across the field.

We used PyCLM to image $\sim 1,000$ MCF10A TagRFP-H2B cells every 30 sec, segment each nucleus, and apply feedback-controlled inputs to drive the red fluorescence of all cells in the field of view to a uniform set point ([Figure 2C](#); [Video S2](#)). Global illumination with a uniform blue light input increased the brightness of all cells ([Figure 2C](#), left versus middle), but cells retained a high degree of variability in fluorescence. By contrast, local feedback control was able to compensate for cells’ intrinsic differences in fluorescence ([Figure 2C](#), right), driving near-uniform

nuclear fluorescence in all cells. We quantified cell-to-cell variability by the coefficient of variation (CV) of cells' nuclear TagRFP intensity under different global conditions (dark versus blue light) and in the presence of feedback control (Figure 2D) as well as by examining the overall distribution of single-cell intensities (Figure 2E). Global illumination increased each cell's fluorescence intensity but had no effect on the CV, whereas feedback control drove a three-fold reduction in CV within 20 min that was maintained for the duration of the control experiment (Figure 2D), leading to a tight distribution of single-cell fluorescent intensities that was peaked at the set point (Figure 2E; Figure S1A). Despite all cells attaining the same final set point, we found that the amount of blue illumination delivered to each cell by the controller scaled inversely with the cell's initial fluorescence (Figures S1B and S1C), indicating that the information about cell-to-cell heterogeneity was now stored within the controller inputs rather than fluorescent output.

Our PyCLM implementation of feedback control is quite general and also enables set points to be chosen locally at different positions in a field of view. We showcased this functionality by providing a grayscale set point image as input, driving the nuclear fluorescence intensity of a field of cells to match the desired intensity pattern (Figure 2F; Video S3). Taken together, these data demonstrate that PyCLM provides a simple, extensible interface to implement intensity-based optogenetic feedback control on large numbers of cells for reducing cell-to-cell variability or controlling cell states.

PyCLM enables precise control over light-delivered developmental signaling patterns

Optogenetics is increasingly used in developmental biology to deliver customized stimulus patterns to embryos and organoids.^{32–37} One problem that is frequently encountered in developmental optogenetics is that individual embryos or organoids may vary in developmental timing, size, or orientation on the microscope. Ideally, it would be possible to scale and rotate optogenetic patterns to deliver matched stimuli across randomly oriented specimens. We developed a PyCLM experimental workflow for developmental optogenetics, focusing on OptoSOS-expressing *Drosophila* embryos as a demonstration system (Figure 2G).³² Local blue light stimulation of the OptoSOS system triggers activation of the Ras/extracellular signal-regulated kinase (ERK) signaling pathway, a required developmental signaling pathway in the early embryo for differentiation of the gut, formation of “head” and “tail” anatomical structures, and patterning the correct number of abdominal segments. We previously showed that local light-activated Ras signaling is sufficient to trigger cell differentiation that leads to tissue movements during gastrulation,³⁸ but these experiments required careful orientation of individual embryos in a fixed light pattern, limiting throughput.

To implement adaptive embryo stimulation, we first developed a new segmentation method to segment embryos instead of cells. In this approach, images of embryos were down-sampled before being passed to the standard CPSAM model. We also wrote a new pattern method to deliver a customizable anterior-posterior light input pattern to each segmented embryo. Our module identifies each segmented embryo's major axis and scales and rotates the light input pattern so that it is delivered

along that axis. This protocol enabled local stimulation at the center of multiple randomly oriented *Drosophila* embryos in a single experiment, producing the expected light-induced developmental perturbations: local changes in membrane-associated tool fluorescence and tissue movement during gastrulation (Figure 2H). Taken together, these results confirm that PyCLM can be used flexibly in optogenetic workflows involving stimulation of cells, tissues, or even whole embryos.

Controlling complex tissue flows using real-time cell segmentation and stimulation

We next applied PyCLM to a canonical use case for optogenetic smart microscopy: delivering local light patterns to control cell movement. Multiple studies have revealed that light-controlled RTKs can serve as potent regulators of cell motility.^{22,39,40} We recently showed that these results also extend to the tissue scale: exposing MCF10A and RPE-1 epithelial cells expressing a light-activated EGFR (OptoEGFR) to static light patterns could produce large-scale epithelial tissue rearrangements.³⁹ Our approach differed from typical optogenetic cell migration experiments, where subcellular light inputs are delivered to each cell to control their directed motion.^{40–44} They also differed from natural RTK stimuli, which are often activated in traveling waves rather than constant patterns.^{18,20} To date, a systematic comparison across these stimulus modes—subcellular versus supracellular inputs and waves at different speeds—has not been performed. We reasoned that PyCLM could be used to deliver each stimulus type, facilitating comparisons between light-induced tissue migration phenotypes.

As a demonstration of light-induced movement, we first considered the problem of using RTK activity to produce rotational tissue flows in an epithelial monolayer, either by delivering subcellular inputs to each cell or sweeping traveling waves across the tissue (Figures 3A and 3B). Rotational or recirculating tissue flows are found in various biological contexts, including hair follicle formation⁴⁵ and vertebrate tailbud elongation,^{46,47} yet their optogenetic control remains elusive. We again used our standard CPSAM segmentation module to identify the position of all cells using their membrane FusionRed fluorescence signal. We then implemented a vortex pattern method that applies a linear gradient of light from the center to the edge of each segmented cell and a rotating_bar method that applies rotating bars of light with a user-specified angular speed, bar width, and periodicity. Blue light patterns were applied once per minute to maintain optogenetic activation, and fluorescent images were collected once every 5 min to monitor cell positions. On our microscope, PyCLM could be parallelized within these time constraints to perform up to fifteen simultaneous experiments on different tissues or fields of view.

We generated a Madine-Darby canine kidney (MDCK) cell line expressing a FusionRed-tagged OptoEGFR receptor, based on the widespread use of MDCK cells as a prime epithelial model system for studying RTK-associated tissue movement.^{19,22,48} We found that both our vortex and rotating-bar stimulus protocols were both capable of producing sustained rotational tissue flows (Figure 3C; Videos S4 and S5). Cell movement was rapidly initiated and maintained throughout each 24 h experiment. We quantified cellular flows in each illumination condition, revealing that subcellular optogenetic stimuli delivered to each cell

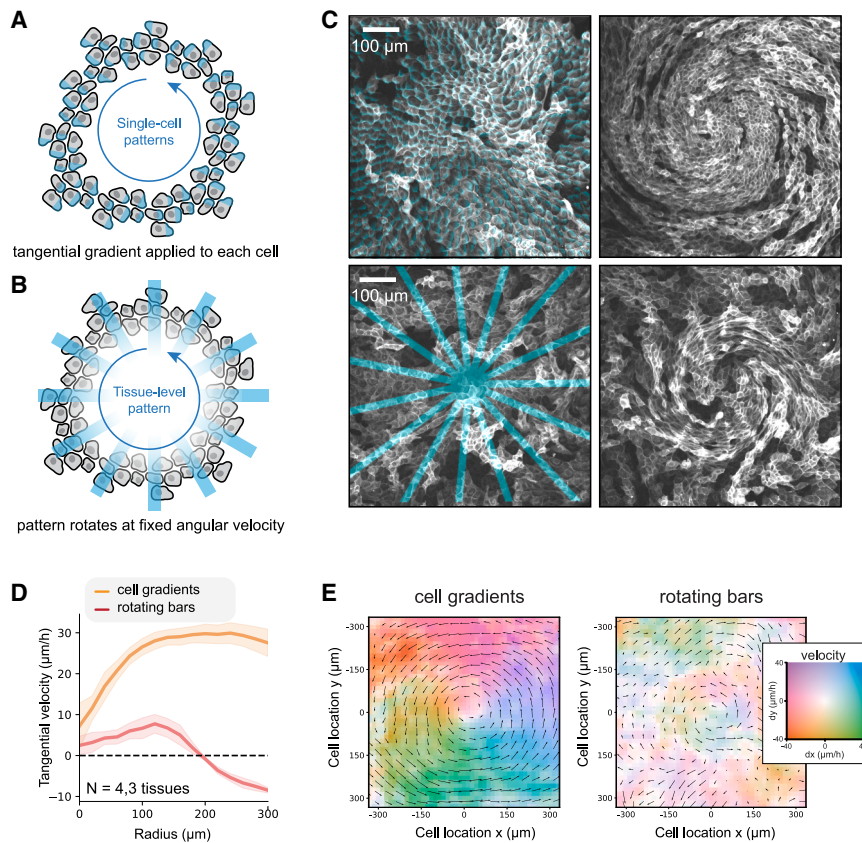


Figure 3. Closed-loop directional stimulation of large cell populations produces user-directed tissue flows

(A and B) We envisioned light stimuli to compare two modes of collective cell movement: single-cell directional stimulation of each cell within a tissue (in A) and collective stimulation using traveling waves of signaling activity delivered to tissue regions as bars of light (in B). The direction of single-cell stimulation and the direction of movement of light bars are tangential with respect to the position relative to the origin.

(C) Left: image of live MDCK OptoEGFR-expressing tissues with the illumination pattern overlaid in teal. Right: tissues imaged after 24 h under the stimulation protocols in (A) and (B).

(D) Quantification of tangential cell velocities as a function of distance from pattern center between $n = 4$ and $n = 3$ tissue replicates, respectively. Lines indicate mean $\pm$ SD of replicates.

(E) Quantification of mean cell velocities as a function of position in the field after 10 h of optogenetic stimulation for each condition. Velocities are shown as a color map (defined in the inset) and as a vector field.

produced more rapid cell movement, with maximum tangential velocities of $\sim 30 \mu\text{m/h}$ compared with $\sim 5 \mu\text{m/h}$ in the case of rotating bars (Figure 3D).

Our analyses also revealed a qualitative difference in tissue responses to the two input types. When optogenetic stimuli were locally delivered to each cell, we observed counterclockwise tissue movement at all positions in the field (Figure 3D, yellow curve). By contrast, delivering rotating bars of light resulted in qualitatively different migration responses that depended on cells' radial positions (Figure 3D, red curve). Cells near the center of the field migrated counterclockwise, but at radial distances greater than $\sim 200 \mu\text{m}$ from the center, migration was reversed and the tissue flowed in a clockwise direction. These differences in migration behavior were also apparent in maps of cell velocities across the field of view (Figure 3E), with rotating bars driving rotational flows in opposite directions at inner and outer positions. Taken together, these data suggest that different dynamic EGFR stimuli—subcellular light patterns versus large-scale bars of light—can produce qualitatively different migration behavior and suggest that varying the dynamics of EGFR activity can even change the overall direction of tissue movement.

The wave speed of EGFR activity controls the direction of epithelial tissue movement

Why might rotating bars produce different directions of migration at inner versus outer positions? One potential explanation is rooted in the observation that a rotating bar's tangential speed varies in proportion to its radial position. Cells at more central positions are stimulated with a slower-moving EGFR wave,

whereas distal positions experience a faster-moving bar. We hypothesized that these differences in wave speed might produce different modes of migration, with slow waves driving migration that tracks the illumination pattern, but fast waves producing cell movement toward the incoming wave. Two *in vivo* systems—zebrafish scale regeneration and mouse epidermal wound healing—exhibit large differences in the speed of traveling RTK activity waves and also differ in the direction of tissue movement (Figure 4A). Zebrafish scales harbor slow FGFR waves with tissue movement tracking the outgoing wave,¹⁸ whereas mouse epithelial tissue harbors fast EGFR waves with tissue movement toward the wave source.^{19,20,22}

Based on these observations, we hypothesized that different wave speeds can switch the direction of cell migration. To directly test this hypothesis, we created a new PyCLM pattern method to deliver linear light waves across a tissue at user-defined speeds. The bar method implements dynamic patterns with three user-defined parameters: the distance between successive bars, the wave speed, and the duty cycle (proportion of space that is illuminated versus dark). We used this method to compare tissue movement in MDCK OptoEGFR monolayers stimulated with "slow" ($15 \mu\text{m/h}$) and "fast" ($60 \mu\text{m/h}$) waves of light. Indeed, we found that traveling waves of illumination produced tissue movement in different directions that depended on the wave speed (Figures 4B and 4C; Videos S6 and S7). Visual inspection of the resulting timecourses revealed that slow and fast waves drove migration in opposing directions, consistent with *in vivo* observations in regenerating zebrafish scales⁴⁰ and wounded mouse epidermis.⁴¹ The dynamics of tissue movement were also qualitatively different depending on wave speed (Videos S6 and S7). In response to slow light waves, cells appeared to move at a consistent speed along the outgoing

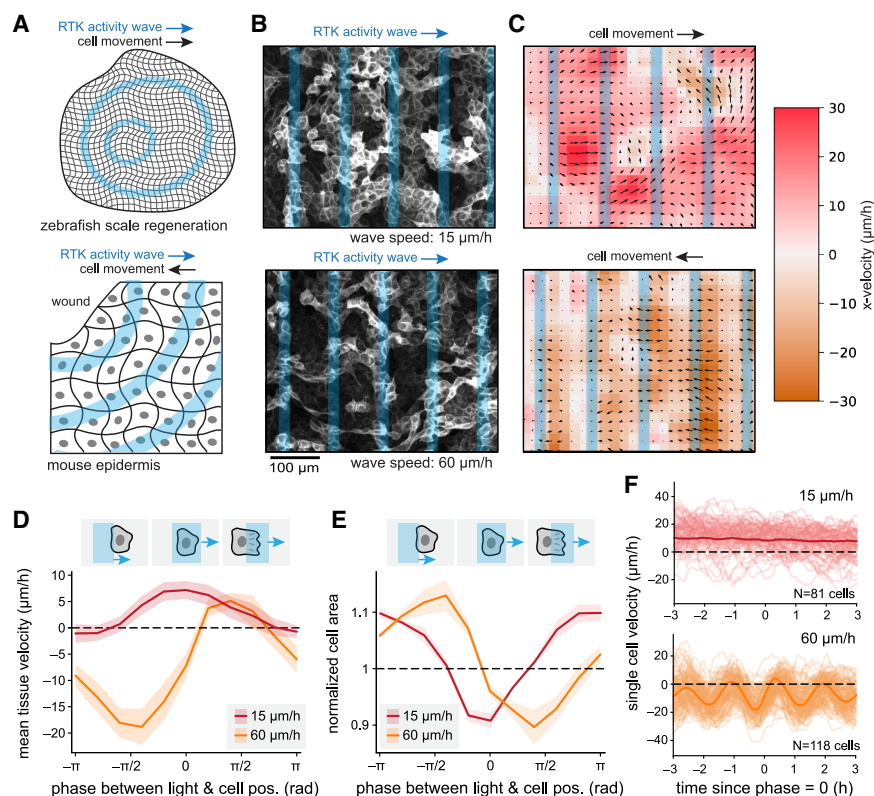


Figure 4. Fast and slow waves of OptoEGFR activity drive qualitatively distinct tissue movement responses

(A) Tissue-scale waves of RTK activity have been observed in mouse epidermal wound healing and zebrafish scale regeneration, where they are thought to control cell movement. Waves propagate radially from a single source (zebrafish: center of scale; mouse: edge of wound). However, tissue moves toward the source of waves in the mouse epidermis and away from the source in zebrafish scale regeneration.

(B) Traveling waves of signaling activity are delivered at horizontal velocities of 15 μm/h (top) and 60 μm/h (bottom) to MDCK cells expressing OptoEGFR.

(C) Quantification of mean cell velocities as a function of position in the field at a representative time point for each stimulation condition. X-velocities are shown as a color map (defined in the inset), and overall velocity is shown as a vector field.

(D and E) Quantification of tissue velocity (in D) and cell area (in E) as a function of the phase of the light wave at each cell's position. Lines indicate mean ± SD from $n = 3$ biological replicates in the 15 and 60 μm/h wave speeds, respectively. Negative phase values indicate the nearest bar of light is approaching the cell, whereas positive values indicate the nearest bar of light is traveling away.

(F) Quantification of single-cell velocities as a function of time since complete overlap by the light bar (defined as phase = 0) for 15 and 60 μm/h light waves. Data show single-cell traces and their mean for $n = 81$, 118 analyzed and tracked cells, respectively.

wave, whereas fast waves drove brief bursts of short-range cell movement toward each incoming wave. Quantification of cell speeds across three biological replicates revealed that tissues moved faster in response to fast waves than slow waves (15 μm/h wave speed: 3.40 ± 1.82 μm/h tissue speed; 60 μm/h: 6.06 ± 1.95 μm/h tissue speed; $p = 0.046$ by paired t test).

To gain further insight into these migration modes, we set out to quantify cell behavior before, during, and after it encounters a wave of light-induced RTK activity. Addressing this question is challenging because the traveling waves are located at different positions in the tissue at each time point, and cells may also move. To account for these effects, we computed the phase of the traveling wave at each time and spatial position, assigning a phase of zero when the light wave is centered on the position. We then plotted the velocity and shapes of all cells as a function of the phase of the light input at their positions (Figures 4D and 4E), observing responses that differed between fast- and slow-wave cases.

For slow waves, we observed the fastest cell speeds for relative phases near zero, corresponding to times when light inputs overlap the cells (Figure 4D; red curve). By contrast, for fast (60 μm/h) waves, we observed rapid cell movement for relative phases near $-\pi/2$, corresponding to the position in space where cells first encounter an incoming light wave (Figure 4D; orange curve). We also noticed that slow and fast waves drove opposing changes to cell geometry (Figure 4E). In the slow-wave case, maximum

cell migration coincided with a reduced cell area, indicating that cells were compressed during movement. By contrast, fast waves produced cell movement during a period of cell expansion. The relative phases between cell movement and cell shape changes are quite different (Figures 4D and 4E), suggesting that these two important variables are likely to respond on different timescales to dynamic RTK stimulation. We also examined the migration behavior of cells over time, computationally synchronizing the cells to a time point when they are centered in the illumination bar at zero phase (Figure 4F). Oscillations in cell speed over time were apparent at 60 μm/h but not 15 μm/h, likely because in the slow-wave case, cells are able to keep up with the outgoing wave and thus only change phase slowly over time.

We also performed analogous traveling-wave experiments in a second cell line, MCF10A cells expressing our OptoEGFR system.^{39,49} MCF10A cells exhibited comparable robust migration in the direction of slow waves, but unlike MDCK cells, they did not exhibit any net displacement in the case of fast light waves (Figure S2A; Videos S8 and S9). Instead, we observed that fast waves led to an oscillatory response, with cells moving equally toward incoming light waves and along outgoing waves (Videos S8 and S9). These results were also reflected in our quantification of cell velocity as a function of light phase (Figures S2B and S2C), which revealed MCF10A responses that were reminiscent of what had been observed in MDCKs (Figures 4D–4F), but with velocities in the fast-wave case that were centered at zero, indicating

symmetric movement with each incoming and outgoing wave. We conclude that some of the light-induced behaviors of epithelial tissues, particularly the directional reversal in response to rapid RTK signaling waves, are dependent on cellular context in ways that are not yet completely understood.

Overall, we find that PyCLM enabled the easy design of experiments that implemented traveling waves of optogenetic stimulation to tissues. These experiments revealed differences between three different modes of cell migration in a canonical epithelial cell context, MDCK cells. Whereas local illumination patterns delivered to single cells produce rapid movement toward stimuli, traveling waves of light produce tissue movement whose direction depends on the wave speed. These data suggest that the different migration behaviors observed in distinct *in vivo* contexts may be related to the characteristic dynamics of RTK signaling waves in each biological context.

DISCUSSION

Smart microscopy is gaining prominence as a transformational technology for cell biology studies. The smart microscopy revolution is driven by numerous recent advances. Artificial intelligence and machine learning make it possible to automate many computational tasks that previously required extensive human intervention, opening the door to real-time cell segmentation, tracking, and analysis. At the same time, an increasing library of biosensors makes it possible to measure ever more traits of the cells being studied. Finally, optogenetic tools provide optical actuators, enabling experimentalists to take advantage of the information gained by imaging to design new perturbations in real-time. Here, we present a software environment for combining each of these modalities to perform dynamic stimulation experiments and automate detection and perturbation experiments on large numbers of cells. Our package is one of a growing number of approaches for performing all-optical smart microscopy experiments that combine fluorescence imaging with optogenetic stimulation.^{12–15} However, PyCLM is relatively unique in emphasis on maximum flexibility—combining multiple experimental workflows in a single multi-point imaging session—with a minimum of programming required by the end user.

Many of the examples we draw upon in this study come from optogenetic control of cell migration. Light-controlled single-cell migration is one of the classic applications of optogenetics in cell biology and indeed was demonstrated in some of the earliest papers in the field.^{41,43} However, these tools have only recently been applied to large-scale tissue movement and collective cell migration,^{22,39} and major questions remain about how mechanical and chemical information is propagated across tissues. Here, we show that PyCLM can perform important classes of experiments for optogenetic studies of collective cell migration: sweeping light inputs across a tissue to recapitulate dynamics observed *in vivo* and locally stimulating single cells at large scales to reveal how the collective polarization of many single cells produces large-scale tissue flows.

We find that both subcellular stimuli and tissue-scale waves of light-activated EGFR activity are sufficient for tissue movement, with more rapid movement produced by subcellular stimuli. We also find that slow (15 $\mu\text{m}/\text{h}$) and fast (60 $\mu\text{m}/\text{h}$) waves of EGFR activity can produce two qualitatively distinct migration behaviors in

a model epithelial context, MDCK cell monolayers. In the case of slow waves, partially illuminated cells follow the outgoing illumination pattern at a relatively constant rate. By contrast, fast waves lead to discrete “ratcheting” movements toward each incoming wave, with rapid and brief bursts of movement triggered by each incoming wave, but limited movement in response to outgoing waves. This ratcheting may be related to differences in cell shape associated with different light phases, producing conditions that are more permissive to movement during incoming waves and restrictive to movement during outgoing waves. These migration data are consistent with the different wave speeds and migration responses in zebrafish scale regeneration and mouse epidermal regeneration: tissue movement that follows $\sim 10 \mu\text{m}/\text{h}$ waves in zebrafish¹⁸ and wound closure that moves toward the source of rapid (100 $\mu\text{m}/\text{h}$) waves in the mouse and in cultured mammalian tissues.^{20,22} Experiments in a second cell line (MCF10A epithelial cells) revealed different dynamics-dependent behaviors: directed motion in response to slow waves but symmetric oscillations in position in response to rapid waves. In future work, it would be interesting to determine the essential features of cellular contexts that elicit these distinct responses.

We further demonstrate that PyCLM is able to perform closed-loop optogenetic stimulation by acquiring images, segmenting biological features of interest, and using these features to compute new light stimuli in real time during experiments. We apply this capability to locally stimulating cell migration, performing feedback control on single-cell fluorescence intensity across populations of more than 1,000 cells, and rotating/scaling developmental signaling patterns for randomly oriented *Drosophila* embryos. One limitation of our current implementation is that it does not track and store information about single cells’ behavior over time. Tracking single cells over time would be useful for certain classes of experiments, such as identifying cells with different dynamic behavior over time (e.g., fast versus slow migrators) or implementing feedback control algorithms that require knowledge of each agent’s history (e.g., integral feedback controllers that integrate accumulated error over time). We aim to add support for cell tracking in future versions of PyCLM, as well as control of multiple light sources to perform still more complex multi-input, multi-output control experiments.

Smart microscopy aims to endow imaging systems with the ability to perform complex experiments without extensive real-time human intervention. Our PyCLM package provides a simple interface for performing one class of smart microscopy experiments: imaging, segmentation, and optogenetic stimulation. Our tool, along with others in this emerging field, will likely enable the next generation of complex imaging-based experiments for biological discovery and control.

RESOURCE AVAILABILITY

Lead contact

Requests for further information and resources and reagents should be directed to and will be fulfilled by the lead contact, Jared Toettcher (toettcher@princeton.edu).

Materials availability

Drosophila strains and mammalian cell lines used in this study are available upon request from the lead contact (toettcher@princeton.edu). No other new materials were generated as a part of this study.

Data and code availability

All data reported in this paper will be shared by the [lead contact](#) upon request. All original code has been deposited at (<https://github.com/Harrison-Oatman/PyCLM>) and is publicly available as of the date of publication. DOIs are listed in the [key resources table](#). Any additional information required to reanalyze the data reported in this paper is available from the [lead contact](#) upon request.

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AUTHOR CONTRIBUTIONS

H.R.O. and J.T. conceptualized the project. H.R.O. and J.H.G. wrote the code. B.C.L., L.N., and J.T. performed all experiments. H.R.O. performed all data analysis. H.R.O. and J.T. wrote the paper with editing from all authors. J.T. supervised the research.

DECLARATION OF INTERESTS

J.T. is a scientific advisor for Prolific Machines and Nereid Therapeutics and is on the advisory board of *Cell Systems*.

STAR★METHODS

Detailed methods are provided in the online version of this paper and include the following:

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SUPPLEMENTAL INFORMATION

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STAR★METHODS

KEY RESOURCES TABLE

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Chemicals, peptides, and recombinant proteins		
Nuclear dye: Hoechst Janelia Fluor 646	Tocris Bioscience	Catalog #6804
Experimental models: Cell lines		
MCF10A clone 5E expressing ErkKTR-iRFP-2A-H2B-RFP	Goglia et al. ²⁹	Available upon request
MDCK cells expressing FusionRed-tagged OptoEGFR	Cell line: this study. OptoEGFR construct: Farahani et al. ⁴⁹	Available upon request
MCF10A clone 5E expressing FusionRed-tagged OptoEGFR	Suh et al. ³⁹	Available upon request
Experimental models: Organisms/strains		
<i>D. melanogaster</i> : UAS-optoSOS	Johnson et al. ³¹	Available upon request
<i>D. melanogaster</i> : 67; histone-miFP	Gift from Shvartsman lab	Available upon request
Software and algorithms		
PyCLM software	This paper	https://doi.org/10.5281/zenodo.20385964 ; https://github.com/Harrison-Oatman/PyCLM
MicroManager Studio Version 2.0.3	Edelstein et al. ¹⁶	Micromanager.org
Cellpose v4.0.6	Pachitariu et al. ²⁷	https://github.com/mouseland/cellpose
Pymmcore-plus v0.13.7	Image Analysis Collaboratory	https://pymmcore-plus.github.io/pymmcore-plus/

EXPERIMENTAL MODEL AND SUBJECT DETAILS

Tissue culture

MCF10A-5E cells⁵⁰ were cultured in DMEM/F12 media supplemented with 5% horse serum (Invitrogen, 16050122), 20 ng/mL EGF (Peprotech, AF-100-15-1MG), 0.5 µg/mL hydrocortisone (Sigma-Aldrich, H0888), 100 ng/mL cholera toxin (Sigma-Aldrich, C8052), 10 µg/mL insulin (Sigma-Aldrich), and 1% penicillin/streptomycin (Gibco, 15140122). MDCK cells were cultured in DMEM low glucose media (Sigma-Aldrich, D5523) supplemented with 1% sodium bicarbonate, 10% fetal bovine serum (R&D Systems, 26140079), and 1% penicillin/streptomycin (Gibco, 15140122). HEK293T cells were cultured in DMEM supplemented with 10% fetal bovine serum (R&D Systems, 26140079) and 1% penicillin/streptomycin (Gibco, 15140122). All cells were maintained at 37°C and 5% CO₂.

Drosophila melanogaster stocks

All stocks were maintained under standard conditions. Embryos expressing OptoSOS and histone-miFP were collected from 67;his3.3-miFP females crossed with UAS-optoSOS males. Embryos were laid in cages kept in the dark at 25°C with apple juice agar plates. Embryos were collected at 0-1h and were not dechorionated before mounting and imaging.

METHOD DETAILS

Cell preparation for microscopy

Cells were imaged on glass-bottomed, black-walled 96-well plates (Cellvis, P96-1.5H-N) coated with fibronectin. Wells of 96-well plates were first incubated with 10 µg/mL fibronectin dissolved in PBS at 37°C for a minimum of 30 min. After fibronectin coating, wells were washed twice with DI water and seeded at 30,000 cells/well 1 day prior to imaging. For improved adhesion, cells were plated into 150 µL of media, then spun down in a tabletop centrifuge for 30 sec. 3 hours prior to imaging, growth medium was replaced with low-growth factor “starvation” medium. Starvation medium was identical to growth medium, but lacked serum and/or growth factors and was supplemented with 3 mg/mL bovine serum albumin. For MCF10A migration experiments (Figure S2), medium was replaced with starvation medium containing 0.1 µM nuclear dye Hoechst Janelia Fluor 646 (Tocris, 6804). All cells were maintained at 37°C and 5% CO₂.

Imaging

All microscopy experiments were performed on a Nikon Eclipse Ti-2 microscope using a Crest V3 spinning disk confocal system, Celesta 545 and 638 nm lasers, a Kinetix 10MP CMOS camera, and a Mightex Polygon 1000 DMD driven by a Sola LED light source. For MCF10A cell experiments in Figure 2, 450 nm light stimuli were applied every 30 sec at 30 mW/cm² for 500 msec; For *Drosophila* embryos in Figure 2 and all stimuli in Figures 3 and 4, 450 nm light stimuli were applied every 1 min at 10 mW/cm² for 500 msec. All images were collected using a 20X air objective lens.

Software PyCLM was written in Python v3.11.5 using pymmcore-plus v0.13.7 and Cellpose²⁷ v4.0.6 for microscope hardware control and cell segmentation, respectively. MicroManager Studio Version 2.0.3 was configured for the microscopy setup described below. All data analysis was performed using custom scripts written in Python. All code and software requirements are available on the project's GitHub page ([www.github.com/Harrison-Oatman/PyCLM](https://github.com/Harrison-Oatman/PyCLM)), with documentation available at readthedocs.io (https://pyclm.readthedocs.io/en/latest/first_time_setup.html). Pymmcore-plus is available at <https://github.com/pymmcore-plus>.

QUANTIFICATION AND STATISTICAL ANALYSIS

Quantification of cell velocities

Cells segmented using Cellpose's CPSAM model²⁷ were tracked frame to frame using a linear assignment problem solver. Velocities for each cell at each time point were calculated as the change in cell centroid divided by the change in time. For line plots in Figure 3D, velocities were binned by distance from the center of the pattern and averaged for each bin for each replicate. Similarly, for line plots in Figures 4D and 4E the velocities were binned by phase and averaged for each bin for each replicate. For both sets of plots (Figures 3D, 4D, and 4E) the line plot shown is the average for each bin over all replicates with error bands representing the standard deviation between replicates.

For heatmaps in Figure 3E, the x-y plane (representing a centered crop of the imaging plane) was divided into a grid. At each grid position, a gaussian kernel ($\sigma = 10\mu\text{m}$) was used to calculate a weighted velocity of the nearby cells from timepoints 9–11h into the experiment. Heatmaps in Figure 4E are calculated similarly, with velocities calculated at the timepoints and field of view shown in Figure 3E, which correspond to 5h after the start of each experiment.

Single cell trajectories in Figure 4F were all taken from a single 15 $\mu\text{m}/\text{h}$ condition and a single 60 $\mu\text{m}/\text{h}$ condition experiment. Trajectories were only included in analysis if they were tracked for at least the period plotted (3 hours before and after a bar of light passed through) in 4F.

Statistical significance in tissue velocities was tested using a paired t-test between three paired replicates of fast (60 $\mu\text{m}/\text{h}$) and slow (15 $\mu\text{m}/\text{h}$) wave speeds.